ERD CASE STUDY 1

Springfield Hospital

The Springfield Hospital (SH) is considering creating a database to keep track of all aspects of the hospital. Impressed by your DB knowledge and skills, the management of SH has decided to hire you to design a conceptual schema first. SH wants to store the information about employees, patients, wards, beds, and treatments in the database.

Each employee has a SSN, name, and salary. For some types of employees, additional information should be recorded. For a doctor, the highest degree obtained and where and when the degree is obtained should be recorded. For a nurse, the level of skill (1 to 5) should be recorded.

Each patient has a SSN, a name, an address (street, city, state, and zip), and several contacts (name and phone number). The patient's disease should also be recorded.

SH has several wards with different functions. Each ward has a number, name, main function, and number of beds.

Each ward has several beds that may contain patients. Beds are numbered within the ward. Two wards could both have a bed 123. The patient will be placed in a bed in a ward relevant to her disease.

Several nurses work in each ward and one of them is in charge of the ward. The nurse in charge of a ward must have a level of at least 4. There must be at least the same number of nurses as the number of beds in any ward.

The patient will be treated by some doctors with various treatments. Each treatment will begin at a specified date and place, and will be associated with a fee. A unique treatment number is assigned to each treatment. A doctor cannot treat more than 20 patients on any single day.